COVER / PROJECT IDENTITY

How can Flipkart reduce cart abandonment and improve conversion?

Programme

PGDM | Big Data Analytics & Database Management | Badruka School of Management

Purpose

Develop a measurable clickstream analytics approach for finding where customers leave the shopping journey and which conversion interventions should be tested.

Evidence statement

Flipkart is a case-study brand. No Flipkart transaction logs have been supplied. This report is an evidence-limited project plan using a documented third-party public e-commerce event dataset.

Submission status

30-page methodological report; empirical outputs, screen captures and experiments are TO BE COMPLETED.

Faculty note: Guidelines specify PostgreSQL; this package uses MySQL by student request, subject to faculty approval.

EXECUTIVE SUMMARY

The challenge is behavioural measurement, not assuming one abandonment cause.

Research problem

A shopper can view a product, place it into a cart, remove it, return later, or buy. An e-commerce team needs comparable session-based measures to prioritize improvements across product discovery, cart review and purchase intent.

Selected external dataset

REES46 publishes free third-party behaviour event logs with view, cart, remove-from-cart and purchase event types. Download and verify the public file before reporting observed performance.

Method

MySQL 8.0 stores the log and runs session-funnel queries. Python profiles fields and creates charts. Databricks shows event transformation into a Delta session table. Power BI displays calculated measures.

Decision output

Hypotheses for checkout design, customer communication and trust improvements, to be evaluated through controlled A/B tests rather than claimed outcomes.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

BUSINESS PROBLEM

What should managers try to understand?

Central question

How can Flipkart reduce cart abandonment and improve conversion?

Management decisions

Where does the measured funnel show attrition? Are there visible product and category differences? Which hypotheses are testable by product managers?

Data needs

At minimum, timestamped events, event type, product/category, user/session identifiers, and documentation of instrumentation.

Unobservable causes

Shipping fee surprise, address-flow issues, payment declines, return policy hesitation and slow checkout cannot be identified from generic view/cart/purchase logs alone.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

OBJECTIVES, SCOPE AND ASSUMPTIONS

Establish the analytical boundary before calculations.

Objectives

1. Audit events and source quality. 2. Define session-based funnel. 3. Count cart sessions without purchase. 4. Analyse product/category/time patterns. 5. Design cross-tool dashboards. 6. Propose an experiment roadmap.

Primary unit

Distinct non-null user_session in the observation period. User-based and event-based statistics must be labelled separately.

Scope

Historical descriptive analytics and test design. Not a real Flipkart operational implementation or audited financial gain estimate.

Assumptions

Tracking identifiers are stable within sessions; event nomenclature is consistent; event timestamp can be parsed as UTC.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

DATASET IDENTITY AND PROVENANCE

Use a source that includes real cart actions.

Publisher

REES46 free ecommerce datasets, with behavior events from third-party online retailers.

Landing URL

https://rees46.com/en/datasets

Dataset access

https://www.kaggle.com/datasets/mkechinov/ecommerce-events-history-in-cosmetics-shop

Evidence availability

The source listing was publicly verified. The CSV was not present in the uploaded project materials and could not be downloaded in this environment. Row count, exact dates and values are unverified locally.

Critical limitation

Third-party cosmetics retail behavior is not necessarily representative of Indian marketplace conditions or Flipkart customers.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

DATA DICTIONARY I

Core event keys and business meaning.

event_time

Timestamp of observed event; normalize UTC and preserve original string.

event_type

view, cart, remove_from_cart, purchase. Defines which activity occurred.

user_session

Source-provided browsing session ID. Primary unit for observed cart/purchase funnel.

user_id

User identifier used for separate repeat-user analyses if privacy, coverage and time span permit.

product_id

Source product key, not a Flipkart SKU. Used for product ranking within the public sample.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

DATA DICTIONARY II

Descriptors and constraints.

category_id

Numeric/string category identifier used for grouping; type needs source inspection.

category_code

Optional human-readable category hierarchy; can have missing values.

brand

Optional brand label; may be missing or inconsistent.

price

Recorded product price; this is not automatically total order revenue or net GMV.

Not in source

Actual checkout-step status, payment gateway outcome, shipping charge, voucher use, marketing channel and user reason for abandoning are not established fields.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

DATA QUALITY PLAN

Data integrity before business inference.

Checks

Raw row count, column count, SHA-256, parsing failures, missing session IDs, source event types, unexpected negative prices and exact duplicates.

Duplicates

Repeated events can be intentional clicks; remove only after verifying tracking semantics. Preserve raw dataset unmodified.

Missing sessions

Exclude rows with missing session key from session calculations; report excluded-event volume and possible bias.

Censoring

Events at the end of the observation window may later convert; avoid claiming that all cart-without-purchase sessions are permanent abandonment.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

WORKFLOW AND ARCHITECTURE

One metric contract across multiple tools.

Stage 1

Download original REES46 CSV -> register source, license, checksum and record shape.

Stage 2

Load raw events to MySQL without silent deduplication. Run validation queries before any aggregation.

Stage 3

Use Python for profiling, session flags, KPI computation and charts.

Stage 4

Use Databricks/Spark to aggregate sessions and persist Delta table when environment permits.

Stage 5

Use Power BI to display matching session metrics. Compare totals with Python and SQL.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

MYSQL IMPLEMENTATION: SCHEMA

Rational DBMS design matching event-level source.

Database

flipkart_conversion_case: dedicated project database, not Flipkart infrastructure.

Table

raw_events stores each source event; BIGINT surrogate event_id is assigned on import. Original event timestamp remains text to prevent silent parsing changes.

Indexes

user_session, event_type and product_id support common groupings and filters.

Validation

COUNT(*), distinct sessions, domain checks, invalid timestamps, missing values and example rows are required before interpreting funnel metrics.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

MYSQL IMPLEMENTATION: IMPORT

DBeaver and loading sequence.

Step 1

Create database and table using sql/01_schema.sql under MySQL 8.0.

Step 2

Import genuine REES46 CSV using DBeaver; map source price to price_text, review quoting, encoding, separator, data types and timestamps.

Step 3

Verify imported record count matches raw CSV and source SHA-256 is recorded.

Step 4

Run sql/02_queries.sql and save output screenshots for the faculty.

Gap

No live MySQL connection or screenshot evidence was supplied; these are implementation instructions, not completed execution.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

SQL: CART FUNNEL

Which carts have no observed purchase in the recorded sessions?

Method

Use MAX(event_type = ...) over each non-null user_session to create view, cart and purchase flags; aggregate flags, not raw event counts.

Primary numerator

Number of sessions that contain cart=1 and purchase=0.

Denominator

Number of sessions with cart=1.

Meaning

A window-dependent cart-without-purchase proxy: absence of recorded purchase is not verified checkout failure.

Expected table

sessions, view_sessions, cart_sessions, purchase_sessions, cart_no_purchase_sessions, cart_no_purchase_pct, session_conversion_pct.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

SQL: PRODUCT AND CATEGORY ANALYSIS

Which parts of the catalog receive cart and purchase activity?

Method

Aggregate cart events by product_id, category_code and brand. Separately report purchase events.

Problem of repeat events

A single user can add the same product repeatedly or purchase multiple items. Raw event ratios cannot substitute for product-level unique session conversion without deduplication rules.

Interpretation rule

High cart-event counts indicate cart interaction, not guaranteed high demand or total sales value.

Evidence

The SQL file includes category, brand, price and product grouping queries; no results are claimed here.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

SQL: TEMPORAL AND WINDOW ANALYSIS

How does purchase activity vary over time?

Daily trends

Group parsed UTC event timestamps to compare view/cart/purchase event volumes.

Session-month cohort

Assign each session to its first observed month and aggregate flags, to avoid counting a session in multiple cohort months.

Window functions

ROW_NUMBER() over session ordered by time shows a diagnostic sequence of tracked events.

Caution

Cross-session purchasing and seasonality interpretation depend on source coverage and instrument quality.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

PYTHON: DATA INSPECTION

pandas workflow for reproducibility.

Read

Import genuine CSV and show dataset shape, dtypes, null profile, unique values and exact duplicate count.

Validate

Assert known event types, non-null session availability and schema. Investigate unexpected values rather than deleting them.

Parse

Convert event_time to timezone-aware datetime using errors=coerce; report invalid counts explicitly.

Output

Audit table and session-level flags. This script is unexecuted until source CSV is placed in the data directory.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

PYTHON: KPI CALCULATIONS

Session-level metrics that can be reconciled across tools.

Cart session count

Count distinct session keys containing >=1 cart event.

Purchase session count

Count distinct session keys containing >=1 purchase event.

Observed cart nonpurchase

Count cart sessions with no purchase event, divided by all observed cart sessions.

View-to-cart proxy

Compare sessions containing a view and sessions containing a cart; note that purchase sessions can be observed without tracked view events.

Validation

Reconcile Python numeric outputs against MySQL SQL results using identical missing-session rules.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

VISUALIZATION I: FUNNEL

Chart plan - real bars appear only after data execution.

Visualization

Three bars: number of observed sessions with view, cart, purchase events.

Business insight it could support

Locates the largest observed change between event stages, though stage populations need not be nested.

Important caution

A session may directly purchase without a tracked cart/view, and these bars are not a perfect linear checkout funnel.

Status

Chart generator is in Python notebook; actual graph pending source CSV execution.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

VISUALIZATION II: DAILY EVENT TRENDS

Time series without fictional results.

Visualization

Daily counts for view, cart and purchase events; units on y-axis are events.

Potential question

Are there high-activity days, abrupt tracking drops, or unusual seasonal spikes?

Diagnostic action

Investigate business promotions and ingestion outages only when external logs or calendars can support the explanation.

Status

Python will generate daily series from the original event timestamps.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

VISUALIZATION III: PRODUCT/CATEGORY

Focus on product-level cart interest.

Visualization

Horizontal bars: top 12 product_id values by cart-event count; separate category plot for recorded purchase events.

Potential question

Which items produce the most observed cart interaction?

Interpretation limit

Cart events do not represent purchased units or profitability. Product names and real Flipkart SKU mapping are not available.

Status

Output conditional on authentic source data and successful notebook execution.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

DATABRICKS: BIG-DATA WORKFLOW

Distributed transforms must be actually run to claim execution.

Load

Spark reads genuine uploaded CSV in Databricks FileStore.

Transform

Group events by user_session, derive view/cart/purchase flags and cart-without-purchase expression.

Verify

Compare raw event count, distinct session count and stage counts against Python/MySQL metrics.

Status

Databricks source script prepared; cluster, workspace execution and result screenshots remain to be completed.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

DELTA LAKE AND GOVERNANCE

Reproducible session table lineage.

Delta write

Persist a session-flags table in Delta format if supported by workspace/permissions.

Evidence

DESCRIBE HISTORY output, table name, Delta version and row count. Do not claim performance improvements without benchmarks.

Governance

Record source checksum, processing rules, table lifecycle and any schema evolution.

Limitations

Delta history and notebook outputs cannot be generated locally without actual Databricks execution.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

POWER BI: DASHBOARD DESIGN

Four or more relevant visuals and three or more KPIs.

Proposed cards

Observed sessions, cart sessions, purchase sessions and observed cart-nonpurchase percent.

Visuals

Session funnel, daily trends, top product cart interactions, category purchase events, and segment slicers where source supports them.

Business value

Shared metric definitions reduce inconsistency across teams.

Evidence required

Actual PBIX or screenshots with data; a DAX design file is not itself a built interactive dashboard.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

POWER BI: DAX AND CONTROLS

Avoid numerator / denominator mismatches.

DAX

Observed Sessions= DISTINCTCOUNT(user_session). Cart Sessions= filtered distinct count; Purchase Sessions similarly.

Rate

Cart No Purchase % = DIVIDE(cart-without-purchase sessions, cart sessions).

Control

Exclude null sessions consistently in imported session metrics; review distinct identifiers after refresh.

Report filters

Date, category and brand only if the session-grain design supports consistent dimensional joins; avoid joining one session to many product rows without controlling duplicates.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

INTEGRATION AND CROSS-CHECKS

Same input; same output; same definitions.

Validation matrix

Source raw row count = MySQL imported row count = Spark raw count, subject to explicit import exclusions.

KPI check

Python session counts must match MySQL result and Delta aggregation using equivalent null handling.

Source trail

Record SHA-256, source link, retrieval date, transformation version and chart export timestamp.

Issue protocol

Any mismatch should be documented as rule -> observed difference -> correction -> verified result, never hidden.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

RESULTS: CURRENT EVIDENCE STATUS

What can be reported today?

Verified

The identified REES46 source family offers free real e-commerce behavior event data and defines event fields, as documented by the publisher.

Prepared

MySQL database, 16 SQL queries, Python notebook, Databricks script, KPI dictionary, BI design, slide deck and methodological report.

Not measured

Dataset records, observed cart nonpurchase rate, session conversion, product rankings, temporal patterns, observed savings, and causal drivers.

Not Flipkart-specific

No Flipkart event-level data, real checkout steps, transaction payment outcomes, or intervention A/B results are supplied.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

RECOMMENDATIONS: HYPOTHESES

Potential interventions - NOT proven dataset findings.

Hypothesis A: clarity

Test transparent delivery charges and time estimates early in the cart journey. Required evidence: exposure and checkout completion events.

Hypothesis B: payment

Test payment retry and payment-options UX with failure-code instrumentation. Current public event data cannot verify payment issues.

Hypothesis C: trust

Test return-window clarity, secure payment messaging and better delivery guarantees, measuring incremental conversion.

Hypothesis D: re-engagement

Test consent-respecting cart reminders against a holdout; avoid treating all saved carts as abandoned.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

EXPERIMENT AND IMPACT MEASUREMENT

How Flipkart could prove impact, if real data were available.

Design

Randomize eligible users/sessions into treatment and control; define exposure, attribution window and deduplication rules.

North-star

Checkout-to-order conversion with pre-registered denominator and guardrail metrics for cancellation, returns and profitability.

Secondary outcomes

Cart-to-purchase conversion, payment success, user complaints, delivery outcomes and incremental contribution margin.

Causal caution

A before/after increase is not automatically caused by the intervention. Seasonal demand and campaigns must be controlled.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

CONCLUSION AND FUTURE WORK

Convert the framework into a credible operational study.

Conclusion

A well-defined session funnel and instrumentation plan are essential prerequisites to diagnosing cart abandonment. The provided scripts make a repeatable starting point.

What cannot be concluded

No actual Flipkart abandonment rate, root cause or improvement is established. Public beauty-retail events represent another retailer and market.

Next steps

Obtain authentic public source; execute notebook/SQL/Delta; build PBIX; reconcile; and, if available, replace proxy analysis with permissioned Flipkart data.

Submission integrity

Label all third-party outputs as proxy benchmarks. Keep the report and presentation aligned with executed evidence.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

TEAM CONTRIBUTION AND LEARNING

Document ownership without inventing personal contributions.

Student identification

To be completed: names and roles of contributors.

Suggested task division

Data provenance/audit, relational SQL, Python EDA, Databricks implementation, Power BI dashboard and report assembly.

Oral demonstration

Each student should explain a metric definition, code execution, data limitation and actionable hypothesis.

Current status

No real team member list or executed contributions were provided; assignment cannot be truthfully completed.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.

REFERENCES AND EVIDENCE REGISTER

Source citations, files and completion checklist.

Primary public dataset

REES46, Free datasets with eCommerce behavior data: https://rees46.com/en/datasets

Kaggle dataset page

https://www.kaggle.com/datasets/mkechinov/ecommerce-events-history-in-cosmetics-shop

Faculty source

Dr. Babu, BSM PGDM Big Data Analytics & DBMS student guidelines (uploaded by student).

Evidence folder

TO BE COMPLETED: original CSV, permission/license, MySQL import and query screenshots, notebook run outputs, Spark/Delta history, PBIX screenshot, GitHub URL.

Academic disclaimer

This is a business-case report, not an official Flipkart analytics report.

Evidence status: project design / public-data benchmark; numeric results await authentic dataset execution.